KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 2nd Year 2nd Term Examination, 2021

ESE 2209

(Bio and Wind Energy Engineering)

Time: 3 Hours. Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

1(a).	What tower height is economical for harvesting wind energy? Discuss.	07
1(b).	Why does wind blow? How would temperature difference make the wind blow?	07
1(c).	How does blade design affect efficiency of wind turbine? Explain the effect of blade pitch on rotor performance of a wind turbine.	08
1(d).	Why angle of attack is so important in wind turbine? With the help of schematic diagram, analyze the blade element theory.	13
2(a).	"Wind turbine is installed at two sites with the same average wind speed may yield entirely different energy output due to differences in the velocity distribution". Justify the statement with example.	08
2(b).	What are the characteristics of Anemometer? Describe the operation of an Anemometer which has no moving parts.	10
2(c).	What are the issues during grid integration in wind power plants? Which procedure is used to integrate wind energy into the grid?	10
2(d).	Are offshore wind farms a good idea? Whether or not? Describe.	07
3(a).	What is the importance of power system stability? Write the classifications of stability.	08
3(b).	An effective dynamic reactive power management system is required in wind power system, why? Describe the functions of static synchronous compensator (STATCOM) to control the reactive power.	12
3(c).	Where in the wind power system should STATCOM be placed and why?	05
3(d).	With a neat sketch, explain the operating regions of the wind turbine.	10
4(a).	Draw the various stages of the life cycle of a wind turbine. Which factors are important during the design of wind turbine?	10
4(b).	Discuss the different ways of calculating the cost of wind energy.	13
4(c).	A wind turbine generates 1576800 kWh and another wind turbine generates 15200 kWh in a year. The generated electricity is sold to the utility at a rate of 5 cents/kWh. The discount rate is 5%. Calculate the present worth of electricity generated by the turbine throughout its life period of 20 years.	07
4(d).	Two wind turbine emitting 104 dB(A) at the source are located 200 m and 240 m away from the residential point, respectively. Calculate the combined acoustic effect of these turbines at the residence. Assume a sound absorption coefficient of 0.005 dB(A)/m and neglect the effect of ambient noise.	05

SECTION – B

5(a).	Why biomass is considered carbon neutral? What are the advantages of biofuel?	10
5(b).	Draw the schematic diagram of a ring matrix pelletizing press.	05
5(c).	How can water be removed from bioethanol? Explain.	07
5(d).	How many steps are there in biogas production? Illustrate the first and second step of biogas production.	13
6(a).	What is COD? How COD of waste water sample can be measured?	12
6(b).	Suppose 15 mL waste water sample is taken to perform a 5 day BOD test that is diluted with dilution water to a total volume of 300 mL when the initial DO concentration is 8 mg/L and after 5 days, has been reduced to 2 mg/L. By determining the BOD of the water sample, what will be the water quality? Discuss.	10
6(c).	Why it is needed to maintain a C/N ratio of 20–30:1 for producing biogas? What is the approximate composition of biogas?	08
6(d).	What is a substrate? Write down the names of three animal biomass sources that can be used as substrate for biogas production.	05
7(a).	Why anaerobic condition is maintained for biogas production? Suppose a house owner has 2 cows and 3 buffalos. Each cow and buffalo is producing dung 10 kg per day and 16 kg per day, respectively. i) By using the dung obtained from the above information in a biogas plant, how much energy can be produced per day? ii) Suppose the house owner has four bulbs of 60 watt and two fans of 80 watt. Is it possible to supply the required electrical energy needed for running the bulbs and fans from the biogas plant if 60% of the energy is lost during the transmission? [Gas production per kg of dung = 0.05 m^3]	10
7(b).	Write down the name of three fast pyrolysis reactor? With a schematic diagram, explain circulating fluidized bed pyrolysis reactor.	10
7(c).	What are components of syngas? In a particular pyrolysis, heating rate is kept at 0.8°C/s. What is the name of this pyrolysis process? Give a brief description of that pyrolysis process.	12
7(d)	What are the factors that affect the gasification process?	03
8(a).	Why is transesterification used in biodiesel production? Write the overall triglyceride transesterification reaction.	07
8(b).	Why is biodiesel considered safe for transportation and human health as compared to conventional diesel fuel? How can biodiesel be used with petroleum diesel fuel?	06
8(c).	Explain the stages involved in bioethanol production process.	12
8(d).	How can temperature and pH affect bioethanol production?	10

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 2nd Year 2nd Term Examination, 2021

EE 2213

(Power Electronics)

Time: 3 Hours. Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

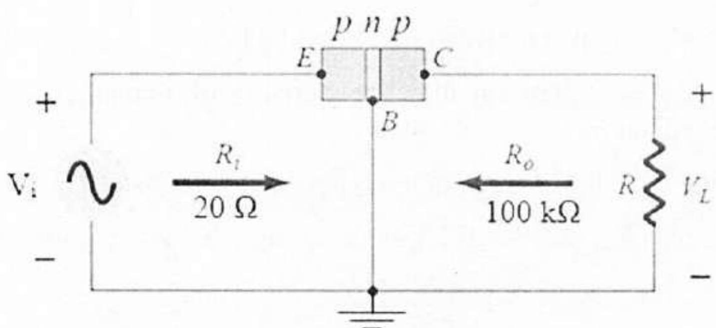
iii) Assume reasonable data if any missing.

SECTION – A

1(a). Write down the necessary conditions for a transistor to work as an amplifier or switch. 05

1(b). Draw and explain the base characteristics and collector characteristics of common emitter configuration. 12

1(c). Calculate the voltage gain (A_v) for the network shown in following figure if V_i = 400 mV and R = 1 kΩ. 07

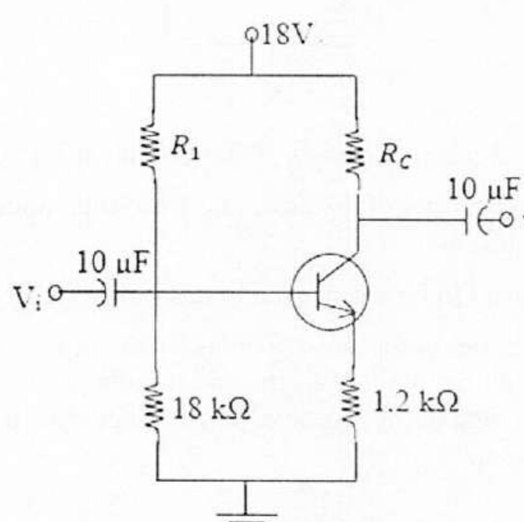


1(d). What is the importance of load line? Explain the effects of changing the levels of collector resistance and V_{CC} on the transistor opperating point (Q) using load line analysis. 11

2(a). What is meant by thermal runaway and how can thermal runaway be minimized? 09

2(b). Derive the equation of stability factor (S) for voltage divider bias configuration and hence show that voltage divider bias configuration is most stable configuration. 15

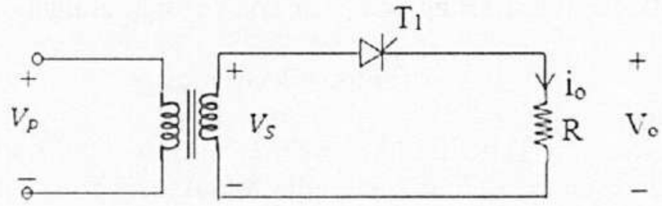
2(c). Given that $I_{CQ} = 2$ mA and $V_{CEQ} = 10$ V, determine R_1 and R_C for the network of following figure. 11



3(a). What is meant by power electronics? Why the study of power electronics is necessary for the students of ESE department? 08

3(b). Explain the operation and draw the necessary waveforms for the speed control of a DC motor by 3-phase full converter system for $\alpha = \frac{\pi}{3}$. 17

3(c). A single-phase halfwave converter has a purely resistive load of R and the delay angle is $\alpha = \frac{\pi}{2}$. Determine the (i) rectification efficiency, (ii) form factor (FF), (iii) ripple factor (RF), and (iv) peak inverse voltage (PIV) of thyristor T_1. 10



4(a). Draw the necessary waveforms for 3-phase bidirectional controller system for $\alpha = \frac{2\pi}{3}$. 12

4(b). Describe the basic principle of single-phase step up cycloconverter. 12

Hence, output frequency = 4 × input frequency.

4(c). Define harmonics and power factor. What are the disadvantages of low power factor? 11

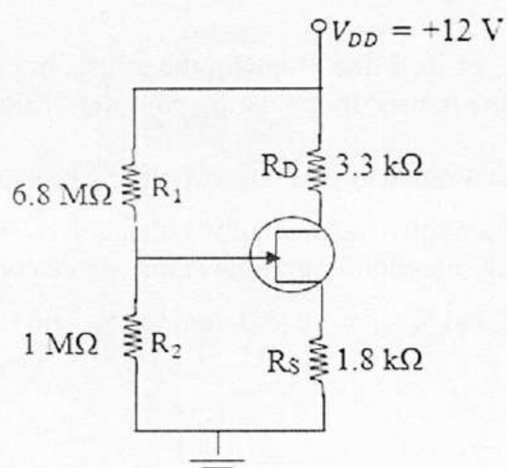
SECTION – B

5(a). What is FET? Distinguish between FETs and BJTs. 07

5(b). Write down the basic operation and characteristics of n-channel depletion type MOSFET with necessary diagram. 13

5(c). What is biasing of JFET? What are the types of JFET biasing? 07

5(d). Determine I_D and V_{GS} for the JFET with voltage divider in following figure, given that $V_D = 7$ V. 08



6(a). Write down the advantages of SCR as a switch in industrial applications. 05

6(b). Draw the two-transistor model of thyristor and from this model, prove that thyristor turns on with a small gate current. 13

6(c) What are the factors need to be considered in designing the thyristor gate control circuits? 07

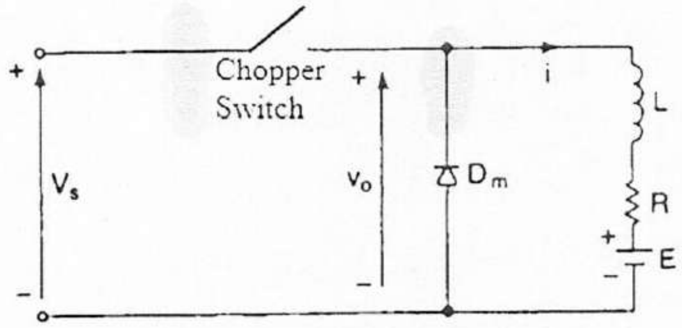
6(d) An SCR half-wave rectifier has a forward breakdown voltage of 150 V when a gate current of 1 mA flows in the gate circuit. If a sinusoidal voltage of 400 V peak is applied, find: (i) firing angle, (ii) average output voltage, (iii) average current for a load resistance of 200 Ω, and (iv) power output. 10

7(a). Write down the differences between DIAC and TRIAC. 08

7(b). Briefly explain the principle operation of step-down chopper and also show that the output voltage can be controlled by varying duty cycle. 12

7(c). Why thyristor is called latching device? 05

7(d).	A chopper is feeding an RL load as shown in figure with $V_s = 220$ V, $R = 5\ \Omega$, $L = 7.5$ mH, $f = 1$ kHz, $k = 0.5$ and $E = 0$ V. Calculate (i) the minimum instantaneous load current, (ii) the maximum peak-to-peak load ripple current.	10



8(a).	Define the following performance parameters of inverter: (i) Harmonic factor of n-th harmonic, (ii) Distortion factor, and (iii) Lowest order harmonic.	09
8(b).	Draw the circuit diagram of 3-phase inverter formed by single-phase inverters.	10
8(c).	Draw a 3-phase inverter with six transistors and six diodes. Also draw its waveforms and equations for (i) phase voltages, and (ii) line voltages for 120 degrees conduction.	16

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
Department of Energy Science and Engineering
B. Sc. Engineering 2nd Year 2nd Term Examination, 2021
Hum 2213
(Economics and Accounting)

Time: 3 Hours. Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

1. Suppose the demand function is $Q_{dx} = 12 - 2P_x$ (P_x is given in dollars), then derive the followings: 35

   a) The individual's demand schedule and individual's demand curve.
   b) What is the maximum quantity this individual will ever demand of commodity x per time period?
   c) What things have been kept constant in the given demand function? Explain.
   d) If there are 10,000 identical individuals of this commodity x, what is the market demand function of commodity x'?
   e) What is law of demand?

2(a). Define individual supply and market supply. What factors affect the supply of any goods or services? 20

2(b). What is neutral equilibrium? If the demand equation is $Q_d = 132 - 8P$ and the supply equation is $Q_s = 6 + 4P$, now 15

   i) Find the equilibrium price P and quantity Q.
   ii) How does a per unit tax " t " affect outcome?
   iii) What is the equilibrium price and quantity if per unit tax t = Tk 4.5?

3(a). Define market. Distinguish between perfect competition and monopoly. 05

3(b). Define short run. Explain the short-run equilibrium of a firm under perfect competition. 20

3(c). Explain income elasticity and cross price elasticity. 10

4(a). What is production? Discuss different factors of production. 15

4(b). What is project evaluation? Explain cost-benefit analysis of a project. 20

SECTION – B

5(a). Define accounting. How accounting helps in making financial decisions by different stakeholders of an organization? 13

5(b). Why it is needed for an engineer to know some accounting? 10

5(c). What is accounting cycle? Describe the steps of accounting cycle. 12

6(a). How do investors use financial statements of an organization before investing or providing loans? 10

6(b). Byron company purchased a drilling machine on January 1, 2015. Related data are given below: 25

Cost = 6,00,000 tk.
Salvage value = 50,000 tk.
Useful life (years) = 5 years
Useful life (hours) = 16,50,000 (machine hours)

Year	Machine hours (used)
2015	4,00,00
2016	3,50,000
2017	3,00,000
2018	3,20,000
2019	2,80,000

Requirements:
i) Show the depreciation schedule of the machine for above five years using straight line depreciation method.
ii) Do the same using units of activity depreciation method.

7(a). Preparation of trial balance is an optional step of accounting cycle. Why do still some organizations prepare trial balance? 10

7(b). Following are the ledger balances of Hashem and Co. Prepare a trial balance on December 31, 2020 with that data. 25

Account titles	**Tk.**
Bank balance	50,000
Prepaid insurance	7,000
Capital	3,00,000
Depreciation	15,000
Land	4,00,000
Salary Payable	22,000
Bills receivable	17,000
Cash	90,000
Accumulated depreciation	75,000
Bad debt	3,000
Sales	1,50,000
Loan payable	1,80,000
Sales discount	2,000
Purchases	70,000
Import duty	4,000
Rent expense	9,000
Supplies	60,000

8.

Nishi Enterprise
Trial Balance
December 31, 2021

Name of the accounts	**Debit (Tk.)**	**Credit (Tk.)**
Cash	25,000	
Prepaid rent	1,400	
Supplies	18,000	
Furniture	32,000	
Accumulated depreciation – Furniture		4,000
Advertising expense	8,000	
Service revenue		80,400
Capital		40,000
Drawing	3,000	
Salary expenses	37,000	
Total	1,24,400	1,24,400

Adjustments:

i) Expired rents 1,000 Tk.
ii) Depreciation on furniture is Tk. 2,000
iii) Unused supplies Tk. 4,000
iv) Accrued salary Tk. 500

Requirements:

a) Prepare a statement of comprehensive income for the year ended December 31, 2021. 20

b) Prepare a statement of financial position as on that date. 15

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 2nd Year 2nd Term Examination, 2021

Math 2213

(Complex Variables and Fourier Analysis)

Time: 3 Hours. Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

1(a). Find all the indicated roots of $\left(-16-16\sqrt{3}i\right)^{1/4}$ and locate them in the complex plane. Also determine the set of points and sketch them graphically in the finite z-plane represented by the inequality $|z+2-2i| \leq |z-2+2i|$. 12

1(b). Prove that the function $f(z)=|z|^2$ is continuous everywhere but no where differentiable except at the origin. 13

1(c). Find the value of a and b such that $f(z)=\cos x\,(\cosh y + a\sinh y) + i\sin x\,(\cosh y + b\sinh y)$ is analytic. 10

2(a). Define a harmonic function. Prove that $u = x^2 - y^2$ and $v = \frac{y}{x^{2^+}y^2}$ are harmonic functions of (x, y), but are not harmonic conjugates. 13

2(b). Evaluate $\int_C z^2 dz$, where C is the 12

(i) arc path $y = x^2$, $1 \leq x \leq 2$, and (ii) straight line path from $(1, 1)$ to $(2, 4)$.

Are they equal? Explain why or why not.

2(c). If an entire function is bounded for all values of z in the finite complex plane, then show that f(z) must be constant. 10

3(a). Determine and classify all the singularities of the function 07

$$f(z) = \frac{z^2}{(z+1)^2}\sin\left(\frac{1}{z-1}\right)$$

3(b). What are meant by simply and multiply connected domain? Using suitable theorem(s), 16

evaluate $\oint_C \frac{3z+5}{z^2+iz}\,dz$, where C is a positively oriented circle $\left|z-\frac{1}{2}\right| = 1$.

Give the statement of those theorem(s) you used.

3(c). Evaluate $\oint_C \frac{e^z - 3z}{(z-1-i)^3}\,dz$; $C: |z-1| = 3$. 12

4(a). Write down the Bessel's function of the first kind of order n. Prove the recurrence relation 10

$J_n'(x) = \frac{n}{x}J_n(x) - J_{n+1}(x)$ and hence show that $J_0'(x) = J_{-1}(x)$.

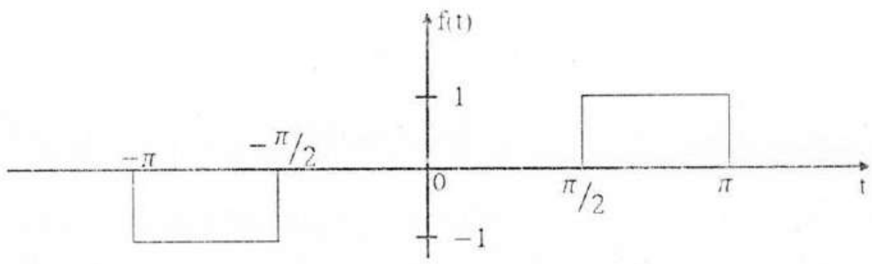
4(b). Show that the Bessel's function of various order can be obtained from the coefficient of different powers of t in the expansion of $e^{\frac{x}{2}\left(t-\frac{1}{t}\right)}$. 13

4(c). Show that $\cos(x\cos\theta) = J_0(x) + 2\sum_{r=1}^{\infty}(-1)^r J_{2r}(x)\cos 2r\theta$ and hence prove that 12

$$J_0(x) = \frac{2}{\pi}\int_0^{\pi/2}\cos(x\cos\theta)\,d\theta.$$

SECTION – B

5(a).	Define periodic function with example. State the assumptions for the expansion of a function in Fourier series.	08
5(b).	Define even and odd functions with example. Obtain Fourier series expansion for the function given by $f(x) = \|\sin x\|$ in the interval $(-\pi, \pi)$.	12
5(c).	Find the Fourier series of the periodic function f(t) represented by the curve shown in following figure.	15



6(a). State Parseval's Identity. By using cosine series for $f(x) = x \ in \ 0 < x < c$, find the sum of the following series $\sum_{n=1}^{\infty} \frac{1}{n^4}$ 12

6(b). Expand the following function in a Fourier series of sine terms only: 12

$$f(x) = \begin{cases} \frac{1}{4} - \frac{1}{x} & 0 < x < \frac{1}{2} \\ x - \frac{3}{4} & \frac{1}{2} < x < 1 \end{cases}$$

6(c). Obtain the complex form of the Fourier series of the following function: 11

$$f(x) = \begin{cases} 0 & -\pi \le x \le 0 \\ 1 & 0 \le x \le \pi \end{cases}$$

7(a). Analyse harmonically the data given below and express y in Fourier series upto the second harmonic. 13

x	0	π/3	2π/3	π	4π/3	5π/3	2π
y	1.0	1.4	1.9	1.7	1.5	1.2	1.0

7(b). Define partial differential equation. Determine the condition for which the following partial differential equation is parabolic and elliptic: 10

$$yu_{xx} - xu_{xy} + yu_{yy} - 3x^2u_y = xe^{-xy}$$

7(c). Form a partial differential equation by eliminating the arbitrary function Φ from $\Phi(x^2 + y^2 + z^2, \ z^2 - 2xy) = 0$. 12

8(a). A bar of length 2 m whose entire surface is insulated including its ends at x = 0 and x = 2 and has initial temperature of the following form: 15

$$2\sin\frac{\pi x}{2} - \sin \pi x + 4\sin 2\pi x, \quad 0 \le x \le 2.$$

Determine the subsequent temperature of the bar where the thermal conductivity of the bar is $k = \frac{1}{4}$.

8(b). A rectangular plate with insulated surface is 10 cm wide and so long compared to its width that it may be considered infinite in length without introducing an appreciable error. If the temperature of the short edge y = 0 is given by 20

$$u = \begin{cases} 20x & 0 \le x \le 5 \\ 20(10 - x) & 5 \le x \le 10 \end{cases}$$

and the two long edges x = 0, x = 10 as well as the other short edge are kept at 0°C.

Determine the temperature u at any point $p(x, y)$.

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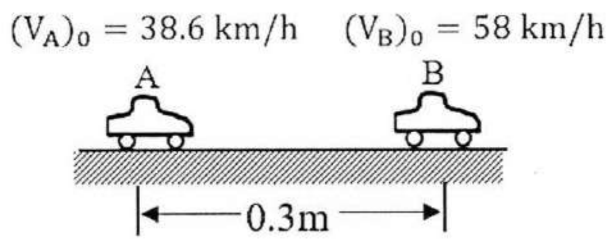
KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
Department of Energy Science and Engineering
B. Sc. Engineering 2nd Year 2nd Term Examination, 2021
ME 2213
(Dynamics and Kinematics of Machineries)

Time: 3 Hours. Full Marks: 210

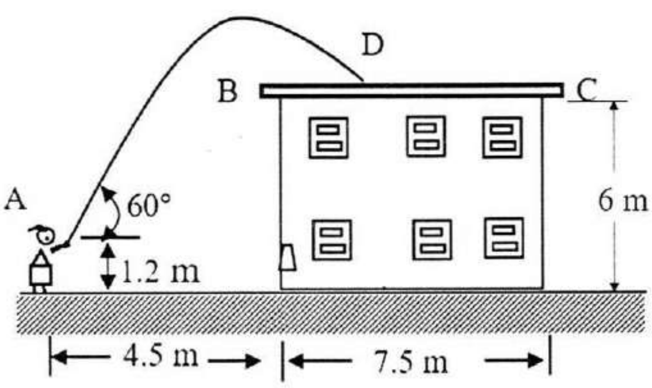
N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

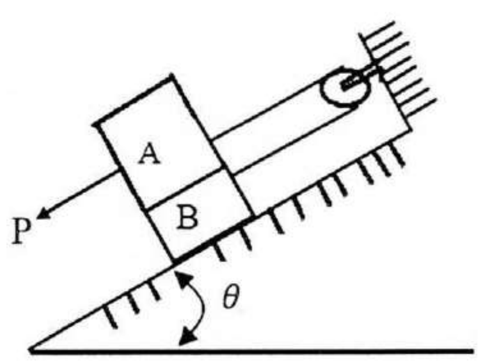
1(a). Automobiles A and B are travelling in adjacent highway lanes and at t = 0 have the positions and speeds shown. Knowing that automobile A has a constant acceleration of 0.55 m/s^2 and that B has a constant deceleration of 0.37 m/s^2, determine (i) when and where A will overtake B, and (ii) the speed of each automobile at that time. 17



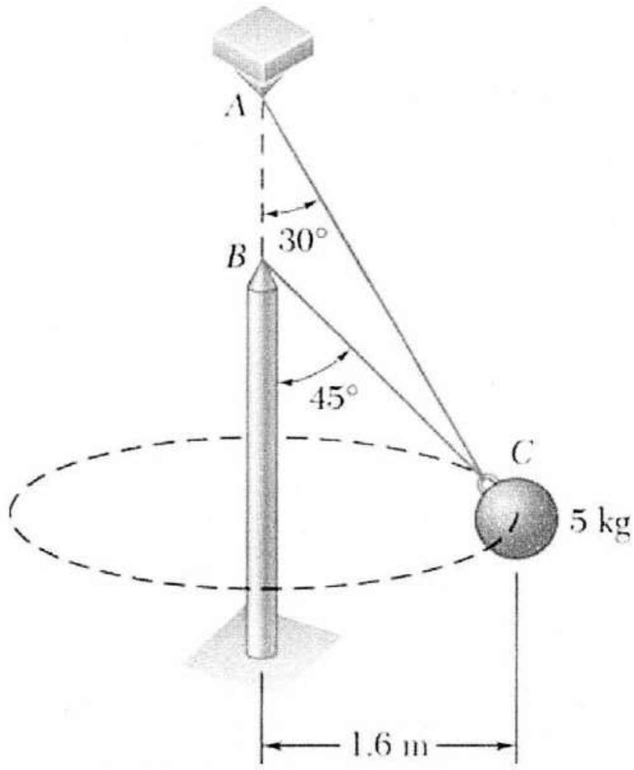
1(b). A nozzle at A discharges water with an initial velocity of 12 m/s at an angle of 60° with the horizontal as shown in figure. Determine, where the stream of water strikes the roof. Check that the stream will clear the edge of the roof. 18



2(a). Block A has a mass of 25 kg and block B has a mass of 15 kg. The co-efficient of friction between all surfaces of contact are $\mu_s = 0.20$ and $\mu_k = 0.15$. Knowing that $\theta = 25°$ and that the magnitude of the force P applied to the block A is 250 N, determine (i) the acceleration of the block A, (ii) the tension in the cord. 18

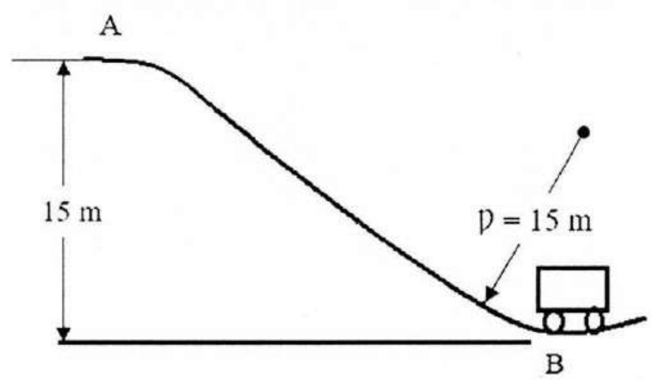


2(b). Two wires AC and BC are tied at C to a sphere which revolves at a constant speed v in the horizontal circle shown. Determine the range of the allowable values of v if the tension in either of the wire is not to exceed 40 N. 17

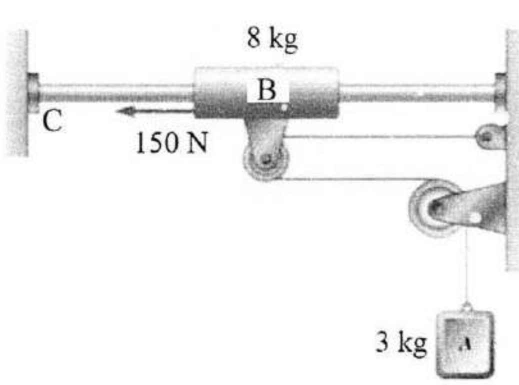


3(a). What does it mean by "coefficient of restitution"? Also, mention the factors that differentiate oblique central impact from direct central impact. 08

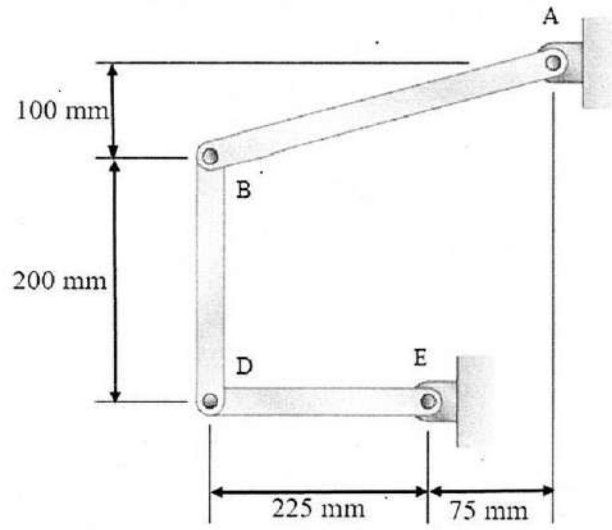
3(b). A roller coaster is released with no velocity at A and rolls down the track as shown. The brakes are suddenly applied as the car passes through point B, causing the wheels of car to slide on the track ($\mu_k = 0.25$). Assuming no energy loss between A and B, and knowing that the radius of curvature of the track at B is 15 m, determine the normal and tangential components of acceleration of the car just after the brakes have been applied. 13



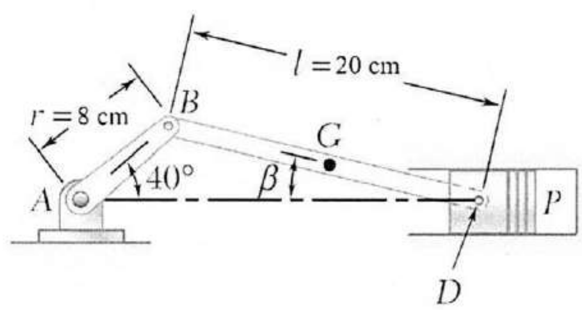
3(c). The system as shown is at rest when a constant 150 N force is applied to collar B. Neglecting the effect of friction, determine, (i) the time at which the velocity of collar B will be 2.5 m/s to the left (ii) the corresponding tension in the cable. 14



4(a). In the position shown in figure, bar AB has a constant angular velocity of 3 rad/s counter clockwise. Determine the angular velocity of bars BD and DE. 18



4(b). In the engine system shown in figure, the crank AB has a constant angular velocity of 2000 r.p.m. For the crank position indicated, using the method of instantaneous center of rotation, determine (i) the angular velocity of the connecting rod BD, (ii) the velocity of the piston P. 17



SECTION – B

5(a). "Flywheel is a mechanical battery"- Explain the statement. 05

5(b). What is the difference in function of a flywheel and a governor? 08

5(c). How maximum fluctuation of energy can be determined with a turning moment diagram for a multi-cylinder engine? 10

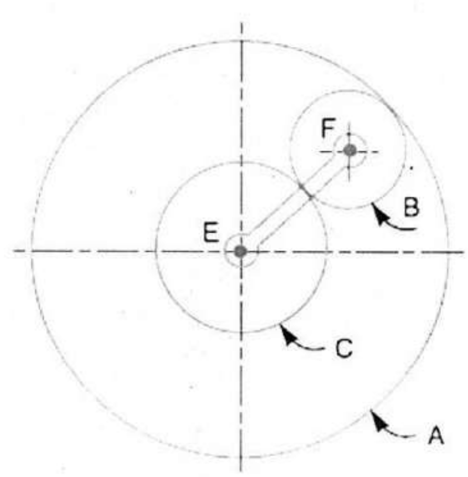
5(d). The mass of flywheel of an engine is 6.5 tonnes and the radius of gyration is 1.8 m. It is found from the turning moment diagram that the fluctuation of energy is 56 kN-m. If the mean speed of the engine is 120 r.p.m., find the maximum and minimum speed. 12

6(a). What are the advantages and disadvantages of gear drive? 08

6(b). Identify the terms typically used in gears with neat sketches. 07

6(c). What are the different types of materials used in gears? 05

6(d). An epicyclic gear train consists of three gears A, B and C as shown. The gear A has 72 internal teeth and gear C has 32 external teeth. The gear B meshes with both A and C and is carried on an arm EF which rotates about the center of A at 18 rpm. If the gear A is fixed, determine the speed of gears B and C. 15



7(a). What is meant by static and dynamic balancing? Describe their significance in terms of balancing of rotating masses. 10

7(b). Explain the statement: "For a multi-cylinder engine, the primary forces may be completely balanced by suitably arranging the crank angles, provided that the number of cranks are not less than four." 06

7(c). Explain the following terms. 09

(i) Tractine force (ii) Swaying couple (ii) Hammer blow

7(d) A single cylinder reciprocating engine has speed 240 r.p.m., stroke 300 mm, mass of reciprocating parts 50 kg, mass of revolving parts at 150 mm radius is 37 kg. If two-third of the reciprocating parts and all the revolving parts are to be balanced, find (i) the balance mass required at a radius of 400 mm, and (ii) the residual unbalance force when the crank has rotated 60° from top dead center. 10

8(a). Draw a 2D sketch of the following governors mentioning unique features of each one. 14

(i) Watt (ii) Porter (iii) Proell (iv) Hartnell (v) Hartung (vi) Wilson–Hartnell

(vi) Pickering

8(b). Determine the relation between the height of a porter governor and the speed of the balls considering frictional force acting on the sleeves. 13

8(c). How does a "Pickering Governor" work? What are its' applications? 08

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